c. Apples and fruits turn brown when cut

Apples and some fruits turn brown due to chemical reaction with oxygen in air. This chemical reaction is called browning. The cells of apples, fruits and other vegetables contain an enzyme called polyphenol oxidase or tyrosinase. When in contact with oxygen it catalyses a biochemical reaction in which the phenolic compounds present in plants become a brown pigment known as melanins.



Fig 10.4 Browning of apple

10.2.2 Environmental Effects

a. Pollution

Our environment provides air to breathe, water to drink and the land to produce food. Due to industrial processes and increasing number of automobiles, our environment is badly affected now-a-days. So, there is an unwanted change in the physical, chemical and biological properties of the environment. This is termed as pollution. The substances which cause these changes are called pollutants. Generally there are three types of pollutions viz air, water and land pollution. Due to increasing human activities, lot of chemical substances are produced artificially which harm all the living and non living things. The types of chemical substances and their effects are given in table below (Table 10.1).



Fig 10.5 Smoke from industries

b. Rusting

What happens to the steel benches and tables during rainy season? They turn into reddish brown. Isn't it? Do you know why? This is because when the iron metal come into contact with water and oxygen, it undergoes a chemical reaction called rusting.



Fig 10.6 Rusted iron barrels

Table 10.1 The types of chemical substances and their effects.

Type of Pollution	Chemical substances	Effects
Air pollution	Carbon dioxide, Carbon monoxide, Oxides of sulphur, Oxides of nitrogen, Chlorofluorocarbons, Methane etc	Acid rain, global warming, respiratory problems etc.
Water pollution	Waste water containing chemical substances Eg. Dyeing industries, Detergents, Oil spillage etc	Decrease in the quality of water, skin diseases etc
Land pollution	Fertilizers like urea, various pesticides, herbicides, solid wastes, plastics etc.	Spoilage of land, cancer, respiratory diseases etc.

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